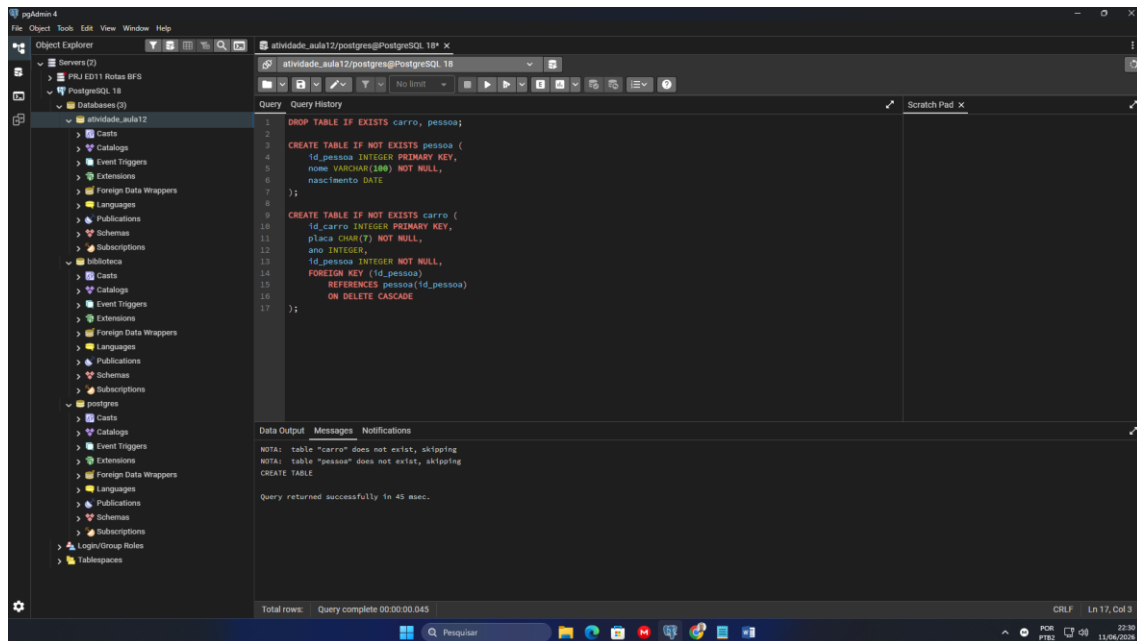


BDR-Atividade_Aula12-ThalesCambráiaDias



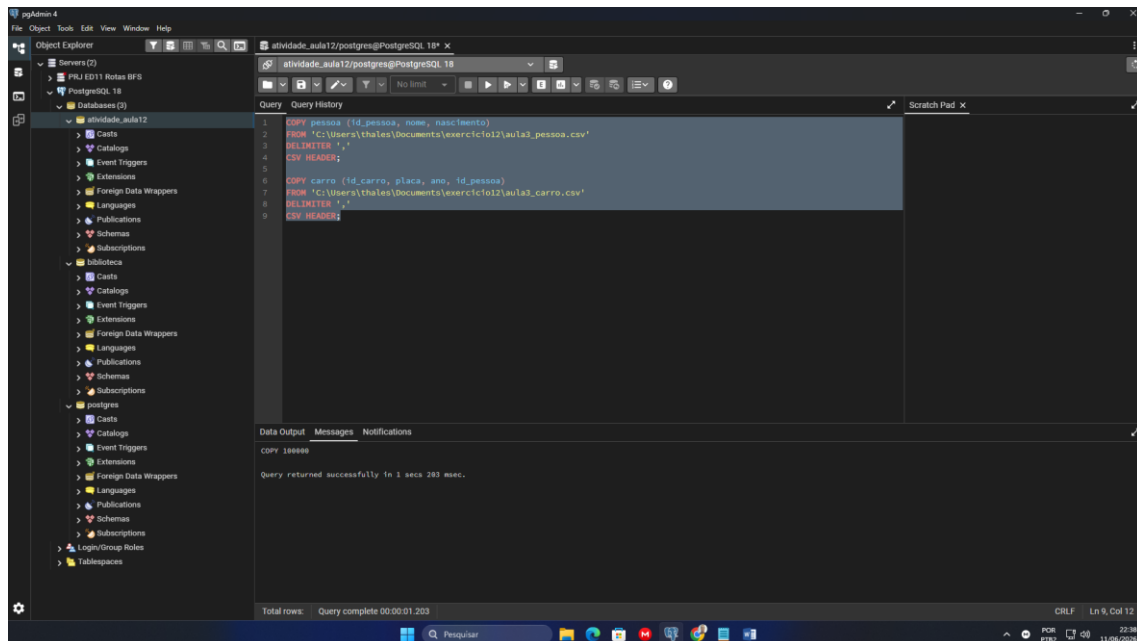
The screenshot shows the pgAdmin 4 interface. On the left, the 'Object Explorer' pane displays the database structure for 'atividade_aula12', including 'Catalogs', 'Event Triggers', 'Extensions', 'Foreign Data Wrappers', 'Languages', 'Publications', 'Schemas', 'Subscriptions', and 'Tables'. The main pane shows the SQL query editor with the following code:

```
1 DROP TABLE IF EXISTS carro, pessoa;
2
3 CREATE TABLE IF NOT EXISTS pessoa (
4     id_pessoa INTEGER PRIMARY KEY,
5     nome VARCHAR(100) NOT NULL,
6     nascimento DATE
7 );
8
9 CREATE TABLE IF NOT EXISTS carro (
10    id_carro INTEGER PRIMARY KEY,
11    placa CHAR(7) NOT NULL,
12    ano INTEGER,
13    id_pessoa INTEGER NOT NULL,
14    FOREIGN KEY (id_pessoa)
15    REFERENCES pessoa(id_pessoa)
16    ON DELETE CASCADE
17 );
```

The 'Data Output' pane at the bottom shows the following messages:

```
NOTA: table "carro" does not exist, skipping
NOTA: table "pessoa" does not exist, skipping
CREATE TABLE
Query returned successfully in 45 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.045'.



The screenshot shows the pgAdmin 4 interface. On the left, the 'Object Explorer' pane displays the database structure for 'atividade_aula12'. The main pane shows the SQL query editor with the following code:

```
1 COPY pessoa (id_pessoa, nome, nascimento)
2 FROM 'C:\Users\thales\Documents\exercico12\aula3_pessoa.csv'
3 DELIMITER ','
4 CSV HEADER;
5
6 COPY carro (id_carro, placa, ano, id_pessoa)
7 FROM 'C:\Users\thales\Documents\exercico12\aula3_carro.csv'
8 DELIMITER ','
9 CSV HEADER;
```

The 'Data Output' pane at the bottom shows the following messages:

```
COPY 100000
Query returned successfully in 1 secs 283 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:01.203'.

pgAdmin 4

Object Explorer

- Servers (2)
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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 EXPLAIN ANALYZE
2 SELECT *
3 FROM pessoa
4 WHERE nome = 'Ana Silva';
5
6 EXPLAIN ANALYZE
7 SELECT *
8 FROM pessoa
9 WHERE nome = 'João Santos';
```

Query History

Scratch Pad

Data Output

Messages

Notifications

Showing rows: 1 to 6 Page No: 1 of 1

QUERY PLAN

text

1	Seq Scan on pessoa (cost=0.00..1838.00 rows=6 width=23) (actual time=2.139..6.252 rows=6 loop=1)
2	Filter (nome = 'João Santos':text)
3	Rows Removed by Filter: 99994
4	Buffers: shared hit=688
5	Planning Time: 0.060 ms
6	Execution Time: 6.253 ms

Total rows: 6 Query complete 00:00:00.045

pgAdmin 4

Object Explorer

- Servers (2)
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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 CREATE INDEX idx_pessoa_nome
2 ON pessoa (nome);
```

Query History

Scratch Pad

Data Output

Messages

Notifications

CREATE INDEX

Query returned successfully in 394 msec.

Total rows: Query complete 00:00:00.394

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 EXPLAIN ANALYZE
2 SELECT *
3 FROM pessoa
4 WHERE nome = 'Ana Silva';
5
6 EXPLAIN ANALYZE
7 SELECT *
8 FROM pessoa
9 WHERE nome = 'João Santos';
```

Query History

Scratch Pad

Data Output

Messages

Notifications

Showing rows: 1 to 10 Page No: 1 of 1

QUERY PLAN

test

1	Bitmap Heap Scan on pessoa (cost=4.34..26.73 rows=6 width=23) (actual time=0.026..0.031 rows=6 loop=1)
2	Recheck Cond: (nome) text = 'João Santos'::text
3	Heap Blocks: exact=6
4	Buffers: shared hit=7 read=1
5	-> Bitmap Index Scan on idx_pessoa_nome (cost=0.00..4.34 rows=6 width=0) (actual time=0.021..0.021 rows=6 loop=1)
6	Index Cond: (nome) text = 'João Santos'::text
7	Index Searches: 1
8	Buffers: shared hit=1 read=1
9	Planning Time: 0.028 ms
10	Execution Time: 0.036 ms

Total rows: 10 Query complete 00:00:00.046

CRLF Ln 9, Col 28

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 DROP INDEX IF EXISTS idx_pessoa_nome;
2
```

Query History

Scratch Pad

Data Output

Messages

Notifications

Showing rows: 1 to 10 Page No: 1 of 1

DROP INDEX

Query returned successfully in 55 msec.

Total rows: Query complete 00:00:00.055

Query returned successfully in 55 msec.

CRLF Ln 2, Col 1

pgAdmin 4

Object Explorer

- Servers (2)
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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 EXPLAIN ANALYZE
2 SELECT *
3 FROM pessoa
4 WHERE nascimento >= DATE '1970-01-01';
5
```

Query History

Scratch Pad

Data Output

Messages

Notifications

Showing rows: 1 to 8 Page No: 1 of 1

QUERY PLAN

text

1	Seq Scan on pessoa (cost=0.00..1938.00 rows=57349 width=23) (actual time=0.028..8.814 rows=57035.00 loops=1)
2	Filter: (nascimento >= '1970-01-01'::date)
3	Rows Removed by Filter: 42965
4	Buffers: shared hit=468
5	Planning:
6	Buffers: shared hit=14
7	Planning Time: 1.775 ms
8	Execution Time: 10.288 ms

Total rows: 8 Query complete 00:00:00.041

CRLF Ln 5, Col 1

22:47 11/06/2028

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 CREATE INDEX idx_pessoa_nascimento
2 ON pessoa (nascimento);
```

Query History

Scratch Pad

Data Output

Messages

Notifications

Showing rows: 1 to 8 Page No: 1 of 1

CREATE INDEX

Query returned successfully in 79 msec.

Total rows: Query complete 00:00:00.079

CRLF Ln 2, Col 24

22:48 11/06/2028

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 EXPLAIN ANALYZE
2 SELECT *
3 FROM pessoa
4 WHERE nascimento >= DATE '1970-01-01';
```

Query History

Scratch Pad

Data Output

Messages

Notifications

Showing rows: 1 to 8 Page No: 1 of 1

QUERY PLAN

text

1	Seq Scan on pessoa (cost=0.00..1938.00 rows=57349 width=23) (actual time=0.010..8.904 rows=57035.00 loops=1)
2	Filter: (nascimento >= '1970-01-01'::date)
3	Rows Removed by Filter: 42965
4	Buffers: shared hit=468
5	Planning:
6	Buffers: shared hit=19 read=1
7	Planning Time: 0.272 ms
8	Execution Time: 10.343 ms

Total rows: 8 Query complete 00:00:00.042

CRLF Ln 3, Col 12

22:48 11/06/2028

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 DROP INDEX IF EXISTS idx_pessoa_nascimento;
```

Query History

Scratch Pad X

Data Output Messages Notifications

DROP INDEX

Query returned successfully in 52 msec.

Total rows: Query complete 00:00:00.052

CRLF Ln 1, Col 44

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 EXPLAIN ANALYZE
2 SELECT *
3 FROM pessoa
4 WHERE nascimento >= DATE '2000-01-01'
5 AND nome = 'Ana Silva';
```

Query History

Scratch Pad X

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

QUERY PLAN

test

1	Seq Scan on pessoa (cost=0.00..2188.00 rows=1 width=23) (actual time=0.004..7.248 rows=2 loop=1)
2	Filter ((nascimento >= '2000-01-01'::date) AND (nome)::text = 'Ana Silva'::text))
3	Rows Removed by Filter: 99998
4	Buffers: shared hit=488
5	Planning:
6	Buffers: shared hit=4 dirtied=1
7	Planning Time: 0.227 ms
8	Execution Time: 7.254 ms

Total rows: 8 Query complete 00:00:00.047

CRLF Ln 5, Col 24

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18

Query

```
1 CREATE INDEX idx_pessoa_nascimento_nome
2 ON pessoa (nascimento, nome);
3
```

Query History

Scratch Pad

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 139 msec.

Total rows: Query complete 00:00:00.141

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18

Query

```
1 EXPLAIN ANALYZE
2 SELECT *
3 FROM pessoa
4 WHERE nascimento >= DATE '2000-01-01'
5 AND nome = 'Ana Silva';
```

Query History

Scratch Pad

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

QUERY PLAN

1	Index Scan using idx_pessoa_nascimento_nome on pessoa (cost=0.42..427.40 rows=1 width=23) (actual time=0.704..1.137 rows=2.00 ms=)
2	Index Cond: (nascimento >= '2000-01-01'::date) AND ((nome) text = Ana Silva::text)
3	Index Searched: 1
4	Buffers: shared hit=11 read=71
5	Planning:
6	Buffers: shared hit=20 read=1
7	Planning Time: 0.368 ms
8	Execution Time: 1.162 ms

Total rows: 8 Query complete 00:00:00.046

PostgreSQL 18

Object Explorer

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Query

```
1 EXPLAIN ANALYZE
2 SELECT *
3 FROM pessoa
4 WHERE nome = 'Ana Silva';
```

Query History

Scratch Pad

Data Output

Messages

Notifications

Showing rows: 1 to 6 Page No: 1 of 1

QUERY PLAN

```
1 Seq Scan on pessoa (cost=0.00..1938.00 rows=6 width=23) (actual time=2.485..6.099 rows=5.00 loop=1)
2 Filter ((nome)::text = Ana Silva::text)
3 Rows Removed by Filter: 99995
4 Buffers: shared hit=488
5 Planning Time: 0.061 ms
6 Execution Time: 6.115 ms
```

Total rows: 6 Query complete 00:00:00.053

PostgreSQL 18

Object Explorer

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Query

```
1 DROP INDEX IF EXISTS idx_pessoa_nascimento_nome;
```

Query History

Scratch Pad

Data Output

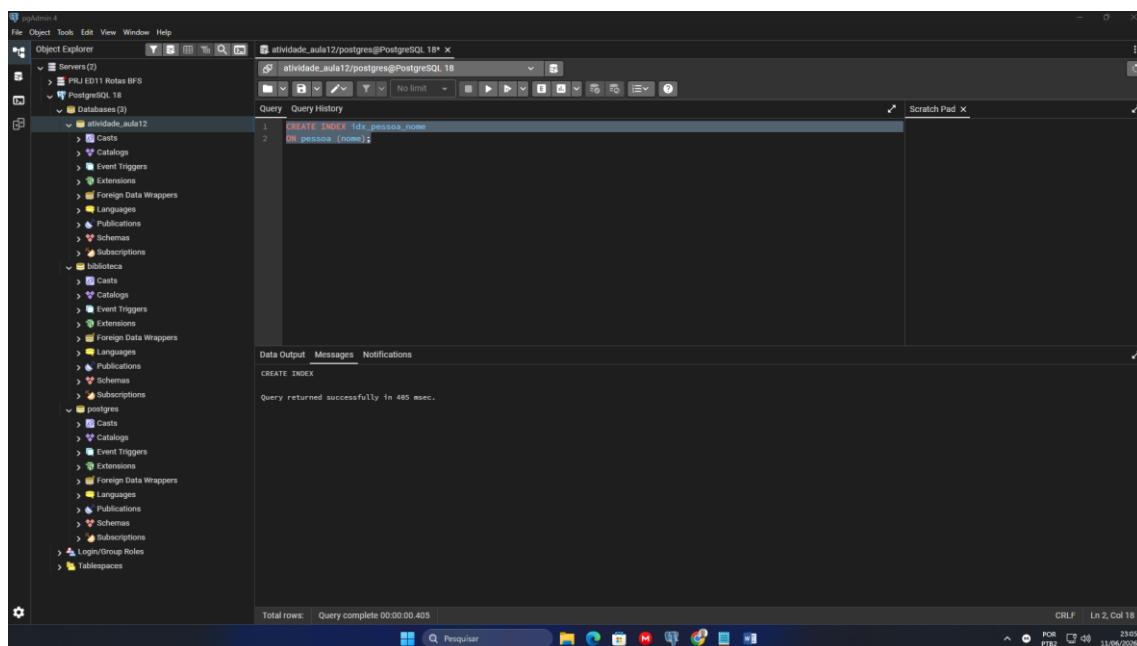
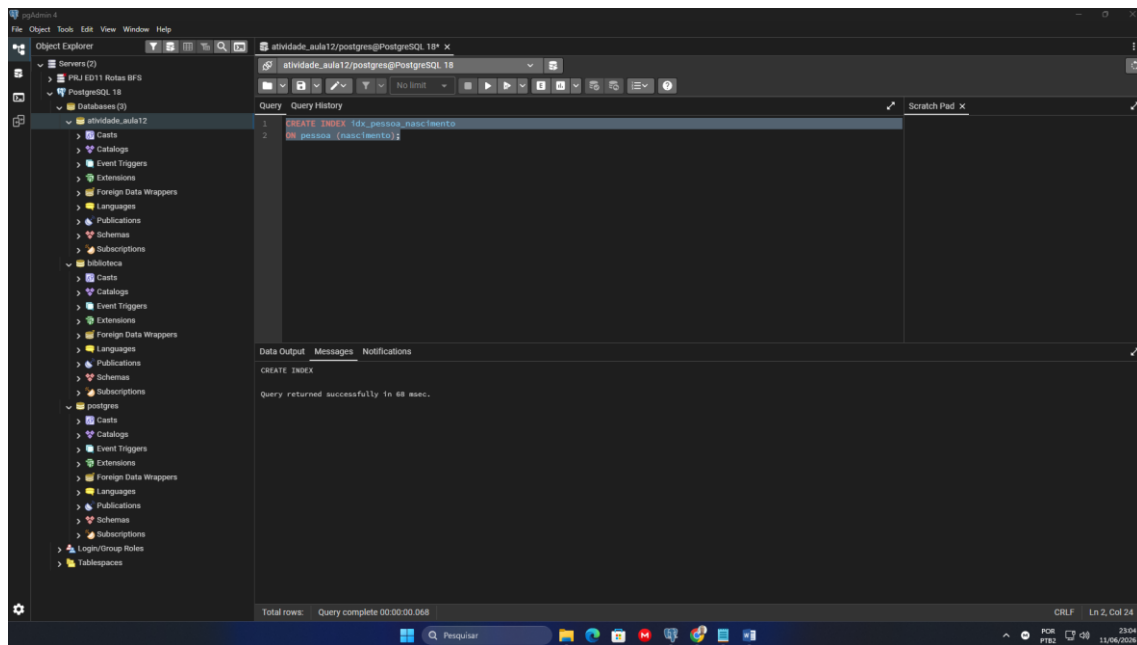
Messages

Notifications

drop index

Query returned successfully in 49 msec.

Total rows: Query complete 00:00:00.049



pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

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 - Languages
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atividade_aula12/postgres@PostgreSQL 18

Query

```
1 EXPLAIN ANALYZE
2 SELECT *
3 FROM pessoa
4 WHERE nascimento >= DATE '2000-01-01'
5 AND nome = 'Ana Silva';
```

Query History

Scratch Pad

Data Output Messages Notifications

Showing rows: 1 to 14 Page No: 1 of 1

QUERY PLAN

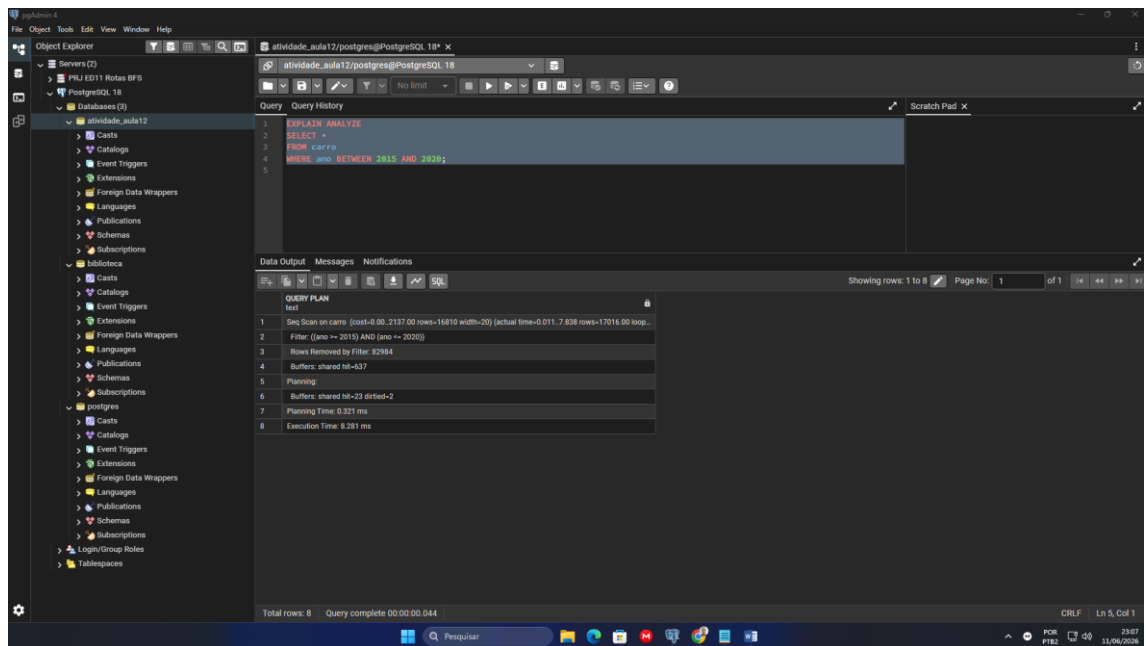
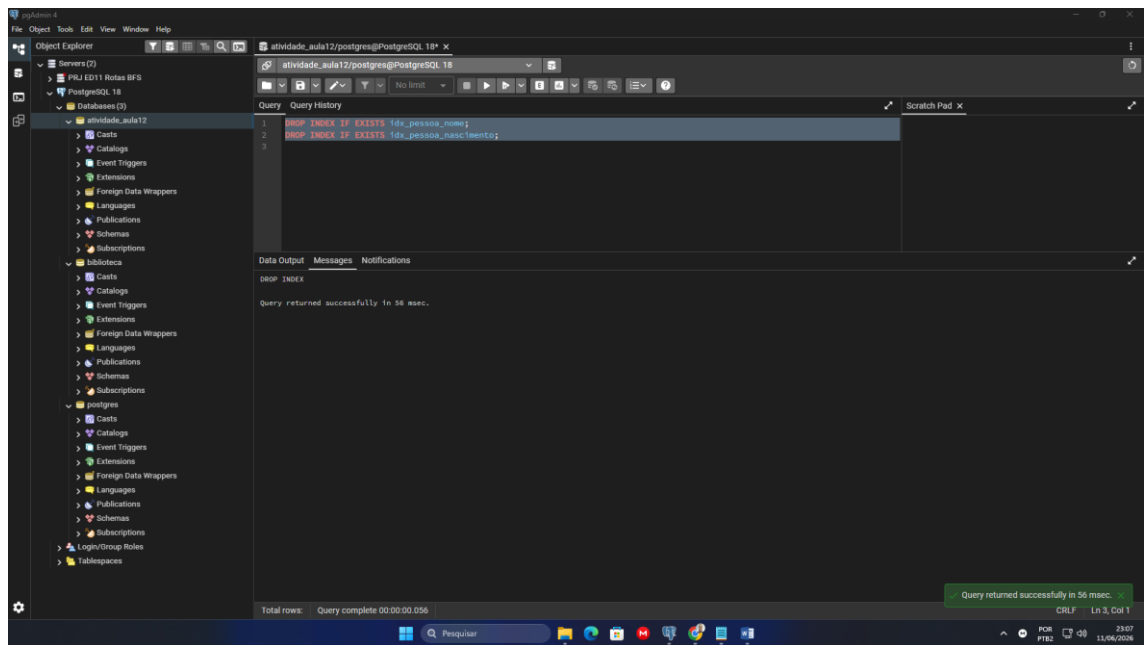
text

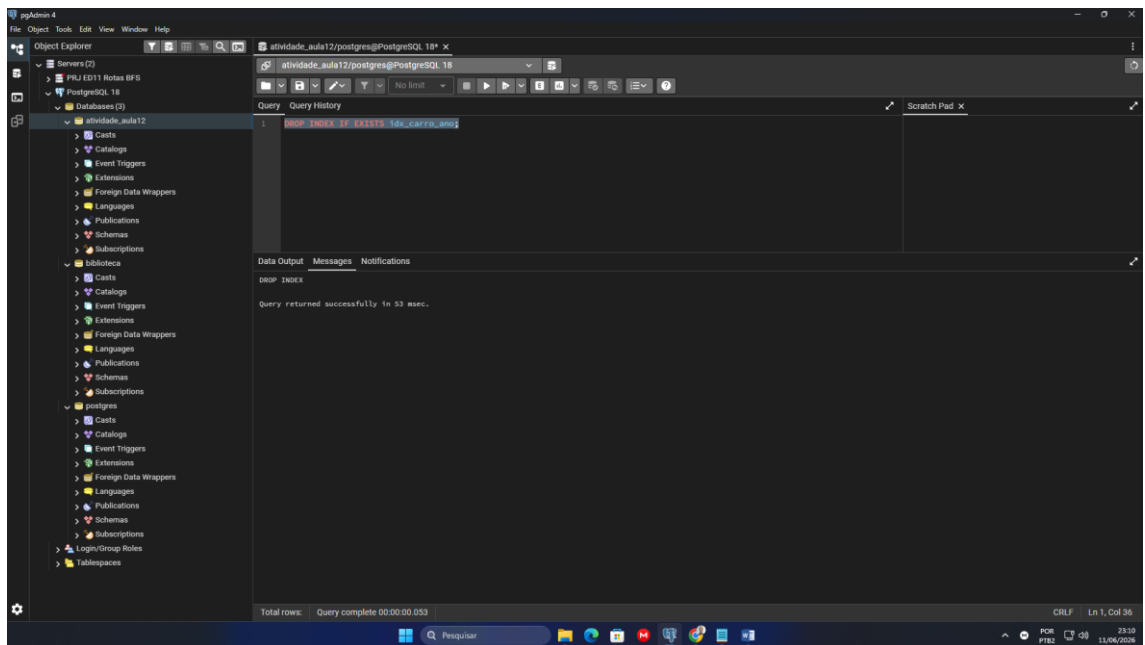
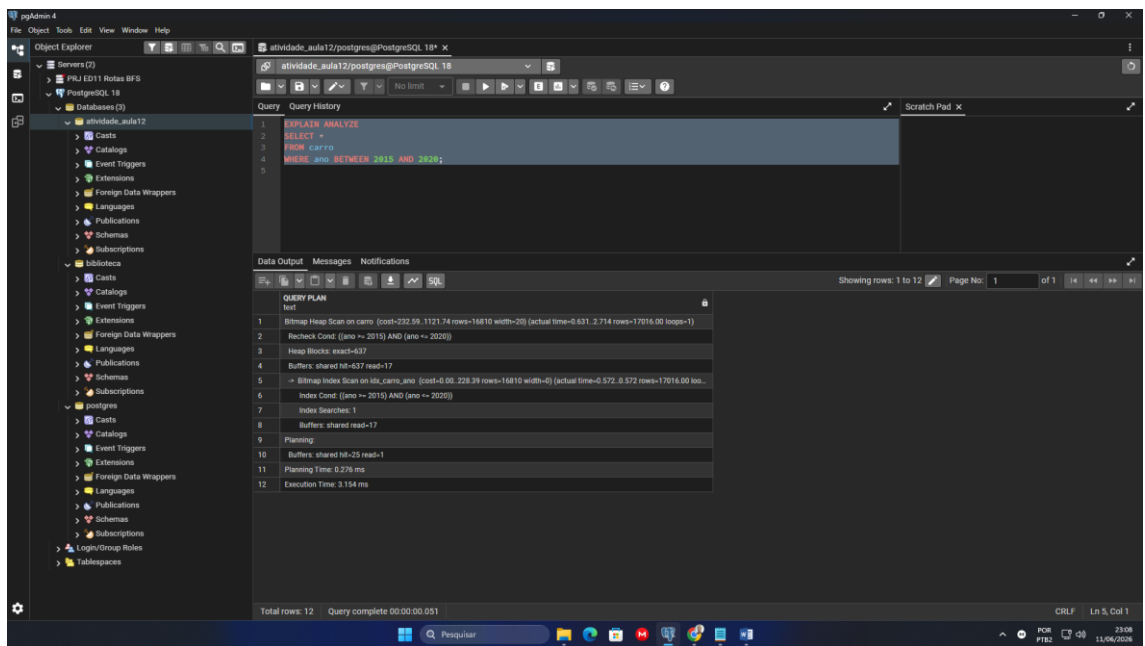
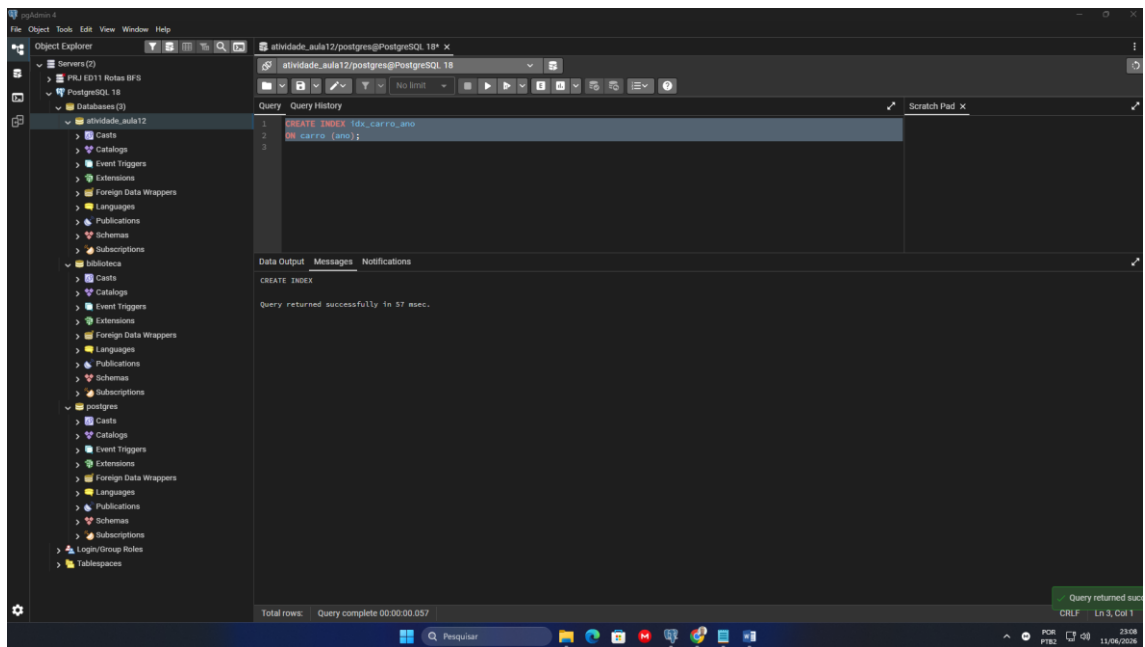
1	Bitmap Heap Scan on pessoa (cost=4.34..26.75 rows=1 width=23) (actual time=0.047..0.051 rows=2.00 loops=1)
2	Recheck Cond: ((nome)::text = 'Ana Silva'::text)
3	Filter: (nascimento >= '2000-01-01'::date)
4	Rows Removed by Filter: 3
5	Heap Blocks: exact=5
6	Buffers: shared hit=5 read=2
7	→ Bitmap Index Scan on idx_pessoa_nome (cost=0.00..4.34 rows=0 width=0) (actual time=0.034..0.034 rows=0.00 loops=1)
8	Index Cond: ((nome)::text = 'Ana Silva'::text)
9	Index Scan: 1
10	Buffers: shared read=2
11	Planning
12	Buffers: shared hit=29 read=2
13	Planning Time: 0.363 ms
14	Execution Time: 0.065 ms

Total rows: 14 Query complete 00:00:00.046

CRLF Ln 5, Col 24

PTB2 23:05 11/06/2026





pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 EXPLAIN ANALYZE
2 SELECT p.nome, c.placa
3 FROM pessoa p
4 JOIN carro c ON p.id_pessoa = c.id_pessoa
5 WHERE p.nome = 'Ana Silva';
```

Query History

Scratch Pad

Data Output

Messages

Notifications

Showing rows: 1 to 16 Page No: 1 of 1

QUERY PLAN

test

1	Hash Join (cost=1938.00..3837.59 rows=6 width=23) (actual time=7.699..18.598 rows=7.00 loops=1)
2	Hash Cond (c.id_pessoa = p.id_pessoa)
3	Buffers: shared hit=1225
4	-> Seq Scan on carro c (cost=0.00..1637.00 rows=100000 width=12) (actual time=0.008..4.354 rows=100000.00 loops=1)
5	Buffers: shared hit=437
6	-> Hash (cost=1938.00..1938.00 rows=6 width=19) (actual time=6.164..6.165 rows=5.00 loops=1)
7	Buffers: 1024 Batches: 1 Memory Usage: 9kB
8	Buffers: shared hit=688
9	-> Seq Scan on pessoa p (cost=0.00..1938.00 rows=6 width=19) (actual time=2.526..6.159 rows=5.00 loops=1)
10	Filter: (nome)=text = 'Ana Silva':text
11	Rows Removed by Filter: 99995
12	Buffers: shared hit=668
13	Planning
14	Buffers: shared hit=40
15	Planning Time: 0.618 ms
16	Execution Time: 18.624 ms

Total rows: 16 Query complete 00:00:00.042

23:19 11/06/2024

pgAdmin 4

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 CREATE INDEX idx_pessoa_nome
2 ON pessoa (nome);
3
4 CREATE INDEX idx_carro_id_pessoa
5 ON carro (id_pessoa);
```

Query History

Scratch Pad

Data Output

Messages

Notifications

CREATE INDEX

Query returned successfully in 446 msec.

Total rows: Query complete 00:00:00.446

23:19 11/06/2024

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 EXPLAIN ANALYZE
2 SELECT p.nome, c.placa
3 FROM pessoa p
4 JOIN carro c ON p.id_pessoa = c.id_pessoa
5 WHERE p.nome = 'Ana Silva';
```

Query History

Scratch Pad

Data Output

Messages

Notifications

Showing rows: 1 to 22 Page No: 1 of 1

QUERY PLAN

text

1 Nested Loop (cost=8.65..98.83 rows=6 width=23) (actual time=0.109..0.205 rows=7.00 loops=1)

2 Buffers: shared hit=17 read=7

3 -> Bitmap Heap Scan on pessoa p (cost=4.34..26.73 rows=6 width=19) (actual time=0.073..0.079 rows=5.00 loops=1)

4 Recheck Cond: (nome)= 'Ana Silva'::text

5 Heap Blocks: exact=5

6 Buffers: shared hit=5 read=2

7 -> Bitmap Index Scan on idx_pessoa_nome (cost=0.00..4.34 rows=6 width=0) (actual time=0.056..0.056 rows=5.00 loops=1)

8 Index Cond: (nome)= 'Ana Silva'::text

9 Index Searches: 1

10 Buffers: shared read=2

11 -> Bitmap Heap Scan on carro c (cost=4.31..12.00 rows=2 width=12) (actual time=0.028..0.021 rows=1.40 loops=5)

12 Recheck Cond: (p.id_pessoa = id_pessoa)

13 Heap Blocks: exact=7

14 Buffers: shared hit=12 read=5

15 -> Bitmap Index Scan on idx_carro_id_pessoa (cost=0.00..4.31 rows=2 width=0) (actual time=0.016..0.016 rows=1.40 loops=5)

16 Index Cond: (id_pessoa = p.id_pessoa)

17 Index Searches: 5

18 Buffers: shared hit=5 read=5

19 Planning

20 Buffers: shared hit=50 read=5 dirtied=3

21 Planning Time: 0.819 ms

22 Execution Time: 0.242 ms

Total rows: 22 Query complete 00:00:00.041

CR/LF Ln 5, Col 28

23/11 11/06/2026

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 DROP INDEX IF EXISTS idx_pessoa_nome;
2 DROP INDEX IF EXISTS idx_carro_id_pessoa;
```

Query History

Scratch Pad

Data Output

Messages

Notifications

drop index

Query returned successfully in 55 msec.

Total rows: Query complete 00:00:00.055

Query returned successfully in 55 msec.

CR/LF Ln 2, Col 42

23/11 11/06/2026

pgAdmin 4

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 CREATE INDEX idx_pessoa_nascimento
2 ON pessoa (nascimento);
3
4 CREATE INDEX idx_carro_ano_id_pessoa
5 ON carro (ano, id_pessoa);
```

Query History

Scratch Pad X

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 149 msec.

Total rows: Query complete 00:00:00.149

CR/LF Ln 5, Col 27

23:17 11/06/2026

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 EXPLAIN ANALYZE
2 SELECT p.nome, c.placa, c.ano
3 FROM pessoa p
4 JOIN carro c ON p.id_pessoa = c.id_pessoa
5 WHERE p.nascimento >= DATE '1988-01-01'
6 AND c.ano >= 2018;
```

Query History

Scratch Pad X

Data Output Messages Notifications

Showing rows: 1 to 26 Page No: 1 of 1

QUERY PLAN

test

1	Hash Join (cost=2760.37..3888.11 rows=8372 width=27) (actual time=13.525..20.703 rows=8500.00 loops=1)
2	Hash Cond (id_pessoa = p.id_pessoa)
3	Hash: shared hit=1302 read=126
4	→ Bitmap Heap Scan on carro c (cost=374.91..1261.28 rows=19950 width=16) (actual time=1.209..3.838 rows=19940.00 loops=1)
5	Recheck Cond: (ano >= 2018)
6	Heap Blocks: exact=437
7	Buffers: shared hit=437 read=56
8	→ Bitmap Index Scan on idx_carro_ano_id_pessoa (cost=0.00..369.92 rows=19950 width=0) (actual time=1.142..1.142 rows=19940.00 loops=1)
9	Index Cond: (ano >= 2018)
10	Index Searches: 1
11	Buffers: shared read=56
12	→ Hash (cost=1838.37..1838.37 rows=42967 width=19) (actual time=12.218..12.219 rows=42787.00 loops=1)
13	Buckets: 65536 Batches: 1 Memory Usage: 271296
14	Buffers: shared hit=463 read=70
15	→ Bitmap Heap Scan on pessoa p (cost=413.29..1838.37 rows=42967 width=19) (actual time=1.826..6.901 rows=42787.00 loops=1)
16	Recheck Cond: (nascimento >= '1988-01-01'::date)
17	Heap Blocks: exact=555
18	Buffers: shared hit=555 read=70
19	→ Bitmap Index Scan on idx_pessoa_nascimento (cost=0.00..502.55 rows=42967 width=0) (actual time=1.761..1.761 rows=42787.00 loops=1)
20	Index Cond: (nascimento >= '1988-01-01'::date)
21	Index Searches: 1
22	Buffers: shared read=70
23	Planning

Total rows: 26 Query complete 00:00:00.053

CR/LF Ln 6, Col 19

23:17 11/06/2026

pgAdmin 4

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atividade_aula12/postgres@PostgreSQL 18*

Query

```
1 DROP INDEX IF EXISTS idx_pessoa_nascimento;
2 DROP INDEX IF EXISTS idx_carro_ano_id_pessoa;
```

Query History

Scratch Pad X

Data Output Messages Notifications

DROP INDEX

Query returned successfully in 56 msec.

Total rows: Query complete 00:00:00.056

CR/LF Ln 2, Col 46

23:18 11/06/2026

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18

Query

```
1 EXPLAIN ANALYZE
2 SELECT *
3 FROM pessoa
4 WHERE nascimento BETWEEN DATE '1988-01-01'
5 AND DATE '1998-12-31';
```

Query History

Scratch Pad

Data Output

Messages

Notifications

Showing rows: 1 to 8 Page No: 1 of 1

QUERY PLAN

Seq

1	Seq Scan on pessoa (cost=0.00:2188.00 rows=15761 width=23) (actual time=0.010:8.266 rows=15745.00 loo...
2	Filter (nascimento >= '1988-01-01'::date) AND (nascimento <= '1998-12-31'::date)
3	Rows Removed by Filter: 84255
4	Buffers: shared hit=468
5	Planning
6	Buffers: shared hit=9
7	Planning Time: 0.179 ms
8	Execution Time: 8.684 ms

Total rows: 8 Query complete 00:00:00.050

Successfully run. Total query runtime: 50 msec. 8 rows affected.

pgAdmin 4

Object Explorer

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atividade_aula12/postgres@PostgreSQL 18

Query

```
1 CREATE EXTENSION IF NOT EXISTS hstore_gist;
```

Query History

Scratch Pad

Data Output

Messages

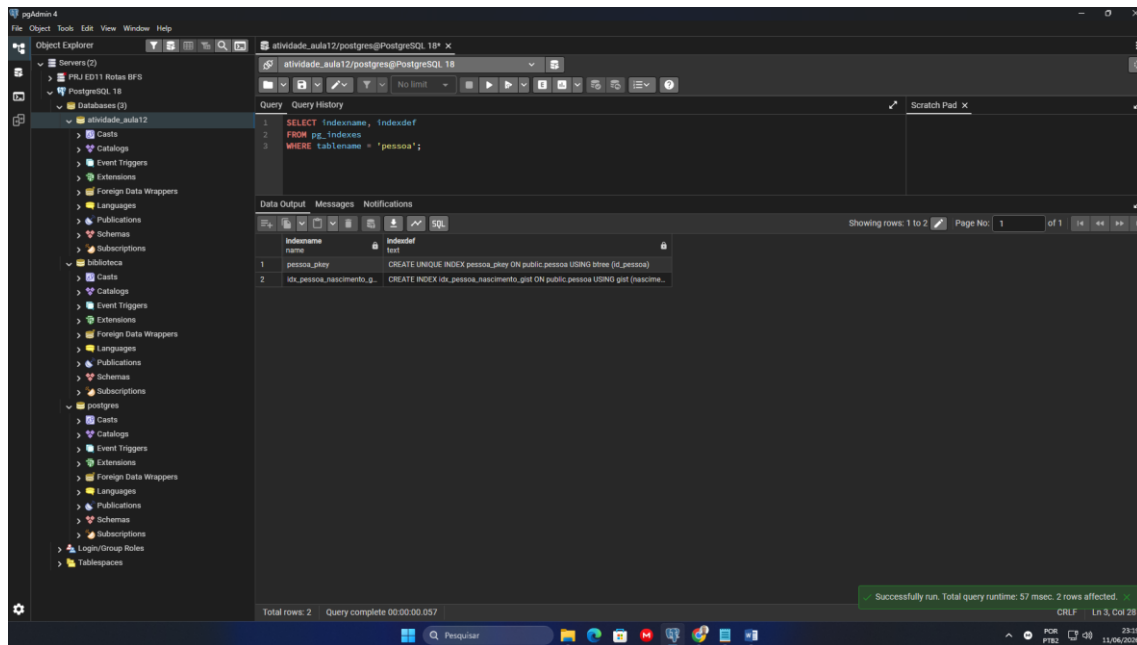
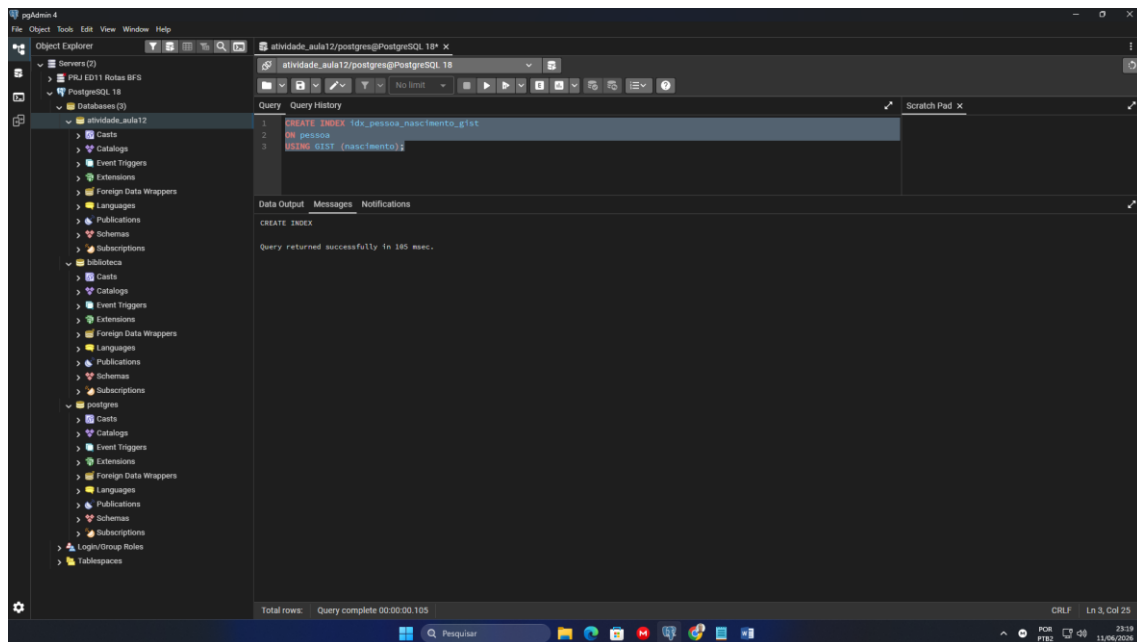
Notifications

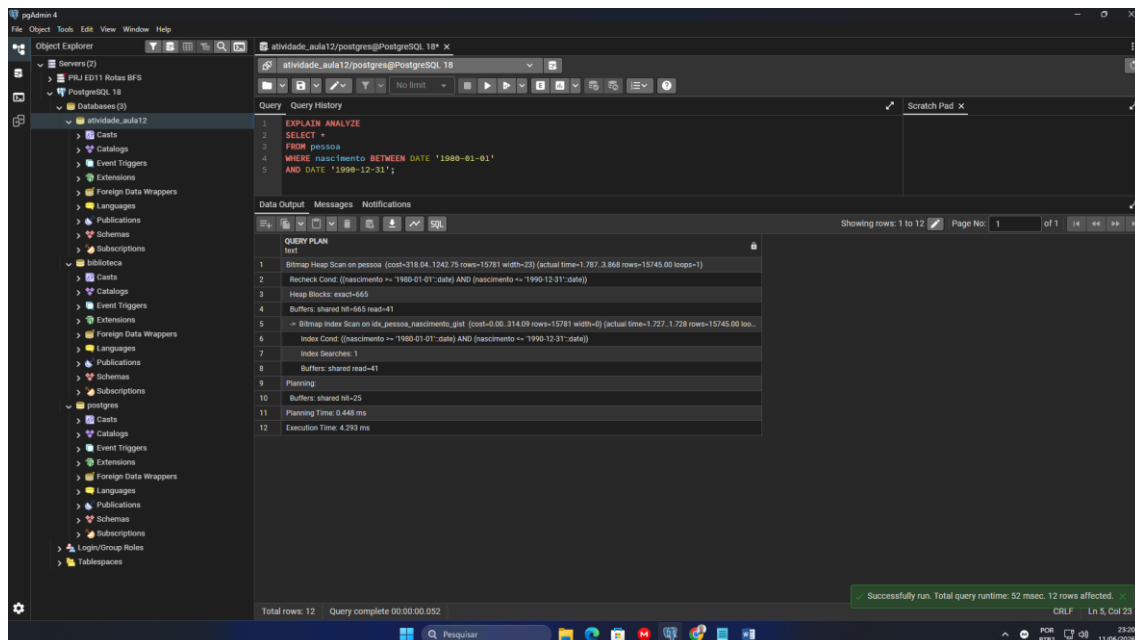
CREATE EXTENSION

Query returned successfully in 189 msec.

Total rows: Query complete 00:00:00.189

Query returned successfully in 189 msec.





Thales Cambraia Dias

Atividade Prática - Aula 12

CRIAÇÃO DE TABELAS

DROP TABLE IF EXISTS carro, pessoa;

CREATE TABLE IF NOT EXISTS pessoa (
 id_pessoa INTEGER PRIMARY KEY,
 nome VARCHAR(100) NOT NULL,
 nascimento DATE
);

CREATE TABLE IF NOT EXISTS carro (
 id_carro INTEGER PRIMARY KEY,
 placa CHAR(7) NOT NULL,
 ano INTEGER,
 id_pessoa INTEGER NOT NULL,
 FOREIGN KEY (id_pessoa)
 REFERENCES pessoa(id_pessoa)
 ON DELETE CASCADE
);

IMPORTAÇÃO DOS CSVs

COPY pessoa (id_pessoa, nome, nascimento)
 FROM 'C:\Users\thales\Documents\exercicio12\aula3_pessoa.csv'
 DELIMITER ','
 CSV HEADER;

COPY carro (id_carro, placa, ano, id_pessoa)
 FROM 'C:\Users\thales\Documents\exercicio12\aula3_carro.csv'

DELIMITER ';' ;
CSV HEADER;

EXERCICIO 1

Parte A – Consulta sem índice explícito

EXPLAIN ANALYZE

SELECT *

FROM pessoa

WHERE nome = 'Ana Silva';

EXPLAIN ANALYZE

SELECT *

FROM pessoa

WHERE nome = 'João Santos';

Parte B – Criação do índice

CREATE INDEX idx_pessoa_nome

ON pessoa (nome);

EXPLAIN ANALYZE

SELECT *

FROM pessoa

WHERE nome = 'Ana Silva';

EXPLAIN ANALYZE

SELECT *

FROM pessoa

WHERE nome = 'João Santos';

Respostas

Antes: Seq Scan

Depois: Index Scan ou Bitmap Index Scan

Sim, normalmente reduz o tempo

O PostgreSQL tende a utilizar o índice quando a busca retorna poucos registros

EXERCICIO 2

DROP INDEX IF EXISTS idx_pessoa_nome;

EXPLAIN ANALYZE

SELECT *

FROM pessoa

WHERE nascimento >= DATE '1970-01-01';

CREATE INDEX idx_pessoa_nascimento

ON pessoa (nascimento);

EXPLAIN ANALYZE

SELECT *

FROM pessoa

WHERE nascimento >= DATE '1970-01-01';

Respostas

Pode utilizar ou não o índice

Nem sempre o plano muda

Se retornar muitos registros a melhoria costuma ser pequena

O otimizador pode preferir Seq Scan quando a consulta retorna grande parte da tabela

EXERCICIO 3

DROP INDEX IF EXISTS idx_pessoa_nascimento;

EXPLAIN ANALYZE

SELECT *

FROM pessoa

WHERE nascimento >= DATE '2000-01-01'

AND nome = 'Ana Silva';

CREATE INDEX idx_pessoa_nascimento_nome

ON pessoa (nascimento, nome);

EXPLAIN ANALYZE

SELECT *

FROM pessoa

WHERE nascimento >= DATE '2000-01-01'

AND nome = 'Ana Silva';

EXPLAIN ANALYZE

SELECT *

FROM pessoa

WHERE nome = 'Ana Silva';

Respostas

Antes: Seq Scan

Depois: Index Scan

Geralmente não usa bem o índice quando filtra apenas nome

A ordem das colunas influencia o aproveitamento do índice

EXERCICIO 4

DROP INDEX IF EXISTS idx_pessoa_nascimento_nome;

CREATE INDEX idx_pessoa_nascimento

ON pessoa (nascimento);

CREATE INDEX idx_pessoa_nome

ON pessoa (nome);

EXPLAIN ANALYZE

SELECT *

FROM pessoa

WHERE nascimento >= DATE '2000-01-01'

AND nome = 'Ana Silva';

Respostas

Sim, normalmente usa os dois índices

BitmapAnd combina os resultados

Filtra pelos dois índices antes de acessar os registros

EXERCICIO 5

DROP INDEX IF EXISTS idx_pessoa_nome;

DROP INDEX IF EXISTS idx_pessoa_nascimento;

EXPLAIN ANALYZE

SELECT *

FROM carro

WHERE ano BETWEEN 2015 AND 2020;

CREATE INDEX idx_carro_ano

ON carro (ano);

EXPLAIN ANALYZE

SELECT *

FROM carro

WHERE ano BETWEEN 2015 AND 2020;

EXERCICIO 6

DROP INDEX IF EXISTS idx_carro_ano;

EXPLAIN ANALYZE

SELECT p.nome, c.placa

FROM pessoa p

JOIN carro c ON p.id_pessoa = c.id_pessoa

WHERE p.nome = 'Ana Silva';

CREATE INDEX idx_pessoa_nome

ON pessoa (nome);

CREATE INDEX idx_carro_id_pessoa

ON carro (id_pessoa);

EXPLAIN ANALYZE

SELECT p.nome, c.placa

FROM pessoa p

JOIN carro c ON p.id_pessoa = c.id_pessoa

WHERE p.nome = 'Ana Silva';

EXERCICIO 7

```
DROP INDEX IF EXISTS idx_pessoa_nome;  
DROP INDEX IF EXISTS idx_carro_id_pessoa;
```

```
CREATE INDEX idx_pessoa_nascimento  
ON pessoa (nascimento);
```

```
CREATE INDEX idx_carro_ano_id_pessoa  
ON carro (ano, id_pessoa);
```

```
EXPLAIN ANALYZE  
SELECT p.nome, c.placa, c.ano  
FROM pessoa p  
JOIN carro c ON p.id_pessoa = c.id_pessoa  
WHERE p.nascimento >= DATE '1980-01-01'  
AND c.ano >= 2018;
```

Justificativa
nascimento participa do filtro
ano participa do filtro
id_pessoa participa do JOIN
índice composto em carro ajuda no filtro e junção

EXERCICIO 8

```
DROP INDEX IF EXISTS idx_pessoa_nascimento;  
DROP INDEX IF EXISTS idx_carro_ano_id_pessoa;
```

```
EXPLAIN ANALYZE  
SELECT *  
FROM pessoa  
WHERE nascimento BETWEEN DATE '1980-01-01'  
AND DATE '1990-12-31';
```

```
CREATE EXTENSION IF NOT EXISTS btree_gist;
```

```
CREATE INDEX idx_pessoa_nascimento_gist  
ON pessoa  
USING GIST (nascimento);
```

```
SELECT indexname, indexdef  
FROM pg_indexes  
WHERE tablename = 'pessoa';
```

```
EXPLAIN ANALYZE  
SELECT *  
FROM pessoa  
WHERE nascimento BETWEEN DATE '1980-01-01'  
AND DATE '1990-12-31';
```

Antes: Seq Scan

Depois: Index Scan ou Bitmap Index Scan

Pode haver melhora dependendo da seletividade

GiST foi utilizado quando considerado vantajoso pelo otimizador